جامعة التقنية والعلوم التطبيقية
University of Technology and Applied Sciences
إبراء Ibra

ACADEMIC YEAR 2025 – 2026

Semester II [February– June]

AI-Powered Virtual Assistant for UTAS Ibra: Improving Student Engagement and Support Services

A PROJECT REPORT

Submitted in partial fulfillment of the requirement for the

Bachelor – Software Engineering

By

UNDER THE GUIDANCE OF

_____

جامعة التقنية والعلوم التطبيقية
University of Technology and Applied Sciences
إبراء Ibra

BONAFIDE CERTIFICATE

This is to certify that the course project entitled **AI-Powered Virtual Assistant for UTAS Ibra: Improving Student Engagement and Support Services** is an approved work done by

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In partial fulfillment for the requirement of the Bachelor – Software Engineering, Computing and Information Sciences Department of the University of Technology and Applied Science – Ibra, for the school year 2025 – 2026.

Course Project Supervisor **Head of Department – IT**

Approved and submitted during the final project presentation held on _____

EXAMINER 1 **EXAMINER 2**

ABSTRACT

The University of Technology and Applied Sciences (UTAS) Ibra is facing challenges in managing student inquiries regarding admissions, course registrations, academic schedules, and technical support. Traditional methods, such as emails, phone calls, and in-person visits, often create inefficiencies and lead to heavy workloads for administrative staff. This research presents the development and implementation of a fully functional AI-powered Virtual Assistant for UTAS Ibra, integrated with the university website. The system provides immediate, 24/7 bilingual support (Arabic/English) to students. The assistant uses Google Gemini AI (gemini-2.5-flash) to understand user intents and create responses. The system provides six main features which include: (1) secure user authentication, (2) real-time access to personal academic data including GPA, class schedules, and attendance records, (3) PDF report generation for timetables, attendance, transcripts, and GPA summaries, (4) a comprehensive admin dashboard for managing FAQs, viewing unanswered questions, and accessing usage analytics, (5) HTML/CSS formatted responses with icons for enhanced user experience, and (6) a 5-star feedback system. The system connects to the university's CIMS database and is accessible via web browser with responsive design. The expected result of this project is to create a system which scales safely and operates easily while improving communication between UTAS Ibra and its students and decreasing administrative tasks by 25 to 50 percent.

ACKNOWLEDGEMENT

We sincerely thank our project supervisor, Dr. Suresh Gowdanakatte Basavarajappa, for his valuable guidance, steady support, and helpful feedback during this project. Their knowledge and encouragement played a key role in shaping our project.

We wish to express our profound gratitude to the administration, staff, and students of the University of Technology and Applied Sciences (UTAS) Ibra for their enthusiastic participation in our surveys and for providing indispensable data and insightful perspectives on the challenges and requirements of the existing support system. Their invaluable input formed the foundation of our empirical investigation.

We also wish to express our appreciation to our colleagues and the Department of Computing and Information Sciences for fostering a supportive learning environment and ensuring access to essential resources. Furthermore, we acknowledge the developers and researchers of open-source tools and libraries, whose innovations enabled the technical advancement of this project.

Finally, we thank our families and friends for their constant support, patience, and encouragement throughout this academic journey. We could not have achieved this without your support.

DECLARATION

Be it known that we.

Bachelor in Software Engineering specialization with a course project proposal entitled AI-Powered Virtual Assistant for UTAS Ibra: Improving Student Engagement and Support Services. Have submitted a complete report. All information in the document has been properly cited, and references have been mentioned in our documentation.

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Chapter I: INTRODUCTION

Background of the Project

The Department of Electronic Systems and Student Services (Student Affairs) at UTAS Ibra is currently facing several challenges, such as inconsistent response times, difficulty in accessing information, and an overwhelmed administrative staff. Students often encounter difficulties like limited office hours and scattered resources, which make it difficult to receive timely assistance for academic inquiries. This project aims to tackle these issues by developing an AI-powered virtual assistant. The virtual assistant will centralize communication and provide consistent support around the clock. Additionally, it will offer bilingual support to ensure both local and international students, effectively bridging the gaps in information access.

Innovative Idea/Concept

This project presents bilingual AI assistants that deliver interactive, real-time assistance, surpassing traditional FAQ pages and static help desks. Its key innovations include bilingual support in English and Arabic, seamless integration with UTAS Ibra's systems, and continuous learning from new questions to improve its responses over time. These features make the assistant a greater and more modern solution compared to existing static information systems.

The system has been developed to deliver five specific features. The first feature involves implementing Google Gemini AI through the gemini-2.5-flash model. The system enables PDF report generation which includes timetable and attendance and transcript and GPA reports. The Admin dashboard provides analytics tools and FAQ management functions. The system enables users to access real-time GPA and schedule and attendance information through CIMS. The system generates HTML and CSS formatted responses which include icon elements. The system uses a 5-star user feedback system to evaluate user satisfaction.

Objectives of the Project

- **General**

  The primary goal of this project is to design and implement an AI-powered virtual assistant that enhances student support services by providing instant, accurate, and bilingual responses for academic inquiries. Designing an AI-powered virtual

assistant that will attend to students' support service needed timely manner and with accurate responses.

- **Specific**

1. Bilingual Capability: Create a Virtual Assistant that understands and responds to queries in both English and Arabic.
2. System Integration: Integrate with the UTAS Ibra website, CIMS and the Student Support System (SSS) [24].
3. Self-Learning Mechanism: Implement a learning mechanism that enables the assistant to enhance its responses based on user interactions.

Effective Evaluation: Assess its effectiveness in reducing administrative workload and improving student satisfaction [2].

Functional and Non Functional Requirements

Functional Requirements:

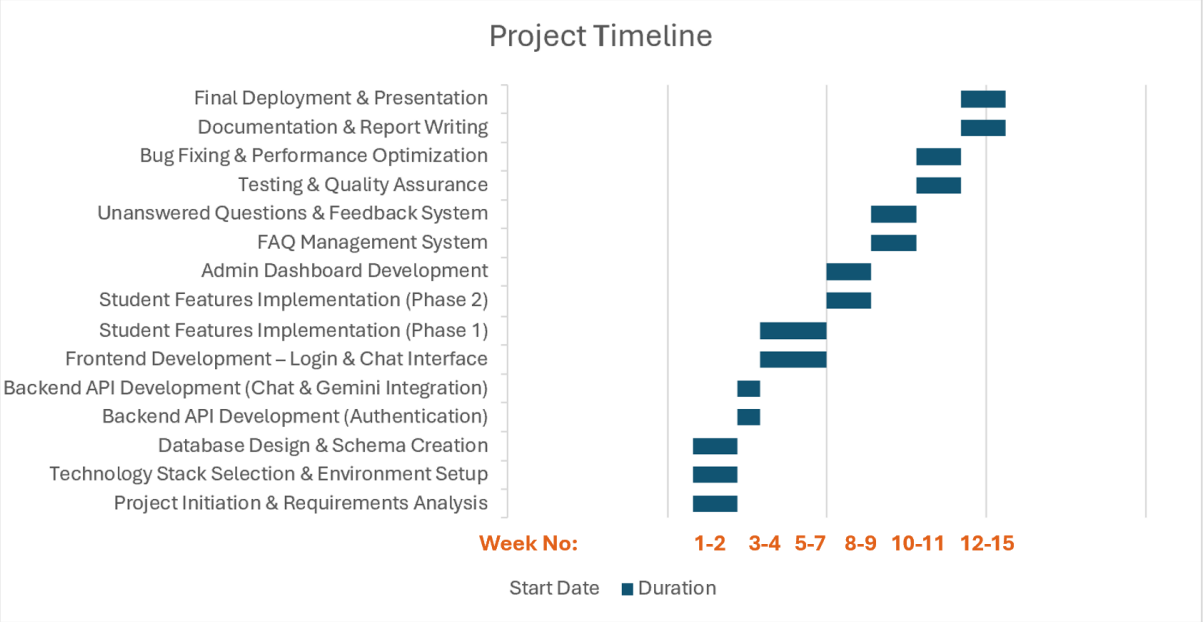
1. HTML develops answers to frequently asked questions through its formatted response system.
2. The system handles both Arabic which uses right-to-left writing and English which uses left-to-right writing.
3. The system retrieves student GPA data and schedule information and attendance records from CIMS system.
4. The system creates PDF reports that include timetable and attendance and transcript and GPA information.
5. The admin dashboard allows users to handle frequently asked questions while they monitor system usage statistics.
6. The system counts unanswered questions through its times_asked tracking feature.
7. The system uses a 5-star feedback system that activates when users log out from the system.
8. The support system operates continuously throughout the entire 24-hour period and throughout all seven days of the week.

Non-Functional Requirements:

1. Performance: Process as much as possible inquiries at the same time.
2. Security: Data encryption and student identity verification when displaying personal data.
3. Usability: Simple and easy to use interface.
4. Scalability: Add mobile applications in the future.
5. Availability: Ensure high system uptime

Project Timeline

S. No.	Week(s)	Task
1	**Week 1-2**	Project Initiation & Requirements Analysis & Technology Stack Selection & Environment Setup & Database Design & Schema Creation
2	**Week 3-4**	Backend API Development (Authentication) & Backend API Development (Chat & Gemini Integration)
3	**Week 5-7**	Frontend Development - Login & Chat Interface & Student Features Implementation (Phase 1)
4	**Week 8-9**	Student Features Implementation (Phase 2) & Admin Dashboard Development
5	**Week 10-11**	FAQ Management System & Unanswered Questions & Feedback System
6	**Week 12-13**	Testing & Quality Assurance & Bug Fixing & Performance Optimization
7	**Week 14-15**	Documentation & Writing Report & Final Deployment & Presentation



Keywords

1. Virtual Assistant
2. Natural Language Processing (NPL)
3. Student Support Services
4. University Services
5. Digital Transformation
6. Artificial Intelligence

Chapter II: Literature Review and Analysis of Related Work

Introduction

This chapter examines existing research on the use of automated systems to support students in universities. The goal is to understand the current state of knowledge, from general trends to specific projects, and to identify the specific weaknesses in our project, the "AI-Powered Virtual Assistant for UTAS Ibra," is designed to solve. While previous studies confirm the value of basic chatbots, they consistently fail to address the need for a system that is fully bilingual [7], deeply connected to university administration, and capable of learning over time. This review will show that our project is not a simple copy of past work, but a necessary and improved solution.

Significance of the Study

The importance of this study is clear and practical. It provides a direct solution to the operational challenges faced by UTAS Ibra [24], aligning with the digital transformation goals of Oman's Vision 2040. For the academic community, it offers an essential real-world example of implementing a bilingual AI assistant in an Omani university context, an area that is currently underrepresented in research [7, 14]. Furthermore, it presents a working model for deep integration with university data systems and a practical self-learning feature, moving behind theoretical discussions to show measurable impact on administrative efficiency and student satisfaction.

Critical Review of Existing Studies

The existing research can be grouped into three main categories, each with special strengths and weaknesses. Broad Reviews and Analyses (such as those by Che Wan Ahmad et al., 2024 [16], and Kuhail et al., 2023 [13]) provide excellent summaries of how chatbots can be used and the common problems they face. However, their wide scope means they cannot provide detailed guidance for specific technical or language needs. Bibliometric Studies (like Chamorro-Atalaya et al., 2023 [5]) are useful for mapping the growth of research in this field but do not offer any practical advice for development. Narrowly Focused Projects (such as Assayed et al.'s 2024 [4] work on academic advising or Okonkwo & Ade-Ibijola's 2021 programming tutor [15]) are effective for their specific tasks but do not offer the university-wide support that UTAS Ibra requires. The Case Study from the University of Sharjah (El Ghandour & Al-Sadi, 2023 [8]) is highly relevant as it proves the value of such a system, but it does not focus on the bilingual challenge or the complexities of deep system integration. A key observation across all this work

is that integration and sophisticated language support are often treated as secondary features, not as primary goals.

Justification for Current Research

It becomes clear after studying the previous research that there are still some important gaps that our study aims to fill. Even though many studies have shown that chatbots are effective, they do not deal with the already complex and multilingual environment at UTAS Ibra [6]. In other words, the reviews by Che Wan Ahmad et al. (2024) and Peyton et al. (2025) talk about the common uses of chatbots but say nothing about creating a strong bilingual feature [5]. Our project solves this by making Arabic-English support a main goal from the very beginning. The studies by Chamorro-Atalaya et al. (2023) and Alkishri et al. (2025) also mention problems in the field, but they stay general, while our project offers a clear and practical system. Moreover, our chatbot is not limited to the simple FAQ model, which Assayed et al. (2024) described as weak, as it connects deeply with both the CIMS and Student Support System at UTAS Ibra to give personalized help. A new feature is the self-learning ability that allows the bot to handle new or unanswered questions [2, 3], a common issue in earlier research, but one that no one has solved yet. Therefore, our project is valid because it not only addresses the problems of language limits, poor integration, and lack of flexibility found in past studies but also provides a more advanced and context-aware solution for UTAS Ibra.

Evaluation of Methodologies

The methods used in the studies we reviewed are quite limited and have several weaknesses. Many of them rely on passive approaches, such as summarizing other papers (systematic reviews) and studying publication patterns (bibliometrics). These methods help to collect and explain existing knowledge but do not create new data or build new systems. The case studies and specific implementation projects provide useful ideas, but their methods are too limited in scope. Our method is stronger because it is complete, practical, and focused on evaluation. We plan to use an applied software engineering approach: a system with the advanced features mentioned in the studies will be built, tested in a real university environment, and its impact will be carefully measured using both numbers (like the time taken to solve inquiries and the number of tickets) and feedback (from surveys and interviews). The entire process, from

building to testing—solves the problems of passive methods and the narrow scope of earlier projects, making sure our research produces both a working tool and reliable results[15].

Comparison of Findings

We are hopeful that our results will not only support but also question and greatly expand the current research. The main reason for our confidence is the common agreement that an AI-powered Virtual Assistant can strongly improve and even double user satisfaction with the help provided. However, we will challenge the common idea in most papers that integration and bilingual support are small issues; our results will likely show that these are the most complex and important factors for success. In addition, our research will make several new contributions to the field. First, we will collect new data on how well Arabic natural language processing performs and how students accept it in a service environment, which will be a major addition to an area that has not been studied much. Second, we will provide clear proof that deep system integration can noticeably reduce administrative work and increase student participation. Finally, the list of unanswered questions will help us build a new and useful feature live map of student concerns and knowledge gaps in the system [2] , giving the university helpful insights that go beyond the chatbot itself. Our research is therefore designed not only to fill the gaps found in earlier studies but also to provide a strong model for the next generation of university virtual assistants.

Ref. No.	Title	Author (Year)	Journal	Type of Journal	Technique / Tool	Key points/Application
1	A review of university chatbots for student support: FAQs and beyond.	Kevin Peyton, Saritha Unnikrishnan, Brian Mulligan (2025)	Spinger-Discover Education	Scopus - Open Access	TensorFlow and PyTorch, Deep Learning Algorithms, Natural Language Generation (NLG), Natural Language Processing, generative techniques	It initially describes chatbot examples related to student administration activities and teaching and learning. Investigates and synthesizes findings from existing university implementations and identify gaps and opportunities for future development. Enhance how information, advice and guidance can be offered to prospective and existing students. 1. Reduce friction for day-to-day activities on campus. 2. Provide Opportunities to enhance Teaching, Learning and Assessment. Following categories: Admissions/Student Recruitment/Administration on Teaching and Learning

2	Application of the Chatbot in University Education: A Bibliometric Analysis.	Chamorro-Atalaya, O.F., et al. (2023)	International Journal of Learning, Teaching and Educational Research *(IJLTER)*	Scopus - Academic Journal	VOSviewer software, Bibliometric analysis, co-word analysis, citation analysis, Systematic Review	This analysis looks at 523 scientific papers from Scopus published between 2013 and 2023. key findings: 1. Research on educational chatbots is growing quickly. 2. The biggest areas of research include intelligent tutoring systems, conversational agents, and automated feedback. 3. Chatbots are being used for: • Helping students • Customizing learning experiences • Automating administrative tasks

3	A systematic review of conversational	Assayed, S. K., Alkhatib, M., &	BUiD Doctoral Research	Springer Conference Proceedings	Systematic review methodology,	Focuses completely on academic advising applications: - Course selection and planning
	AI chatbots in academic advising.	Shaalan, K. (2024)	Conference 2023 (Springer Cham)		NLP analysis framework, chatbot evaluation metrics	- Degree progress tracking - Academic policy information - Career pathway guidance Specific applications include: - Automated degree audit systems - Personalized course recommendations - Progress monitoring alerts

4	Conversational AI in Education: A General Review of Chatbot Technologies and Challenges.	Alkishri, W., Yousif, J. H., & Al Husaini, Y. (2025)	[Journal Name to be Verified] (Pre-publication / Journal to be verified)	Academic Journal / Research Repository	Natural Language Understanding (NLU), dialog management systems, machine learning algorithms	Provides a full overview of technology: Rule-based versus AI-powered systems, focused recognition techniques, and context management challenges. It documents continuing issues such as handling unclear queries, maintaining conversation context, limitations in Arabic language processing, and scalability concerns. While it identifies problems, it does not offer implementation solutions.
5	Interacting with educational chatbots: a	Kuhail, M. A., Alturki, N., Alramlawi, S.,	Education and Information	Scopus - Academic Journal	Human-computer interaction analysis, usability	The analysis covers 62 studies on user interaction patterns, focusing on several key areas:

	systematic review.	& Alhejori, K. (2023)	Technologies (*Springer)*	Scopus & Web-of-Science Indexed, Peer-Reviewed Journal	testing frameworks, user experience evaluation, Systematic Review	- Preferred communication styles - Responses to time expectations - Personality and tone preferences Key findings include: - Students prefer informal and friendly interactions. - Quick responses are crucial for user satisfaction. - Visual elements enhance engagement. - Personalization increases adoption rates.

6	Unleashing the Potential of Chatbots in Education: A State-Of-The-Art Analysis.	Winkler, R., & Söllner, M. (2018)	Academy of Management Annual Meeting (AOM)	Conference Proceeding	Early NLP systems, rule-based architectures, basic machine learning	Foundational analysis covering the historical development of educational chatbots, classification of chatbot architecture, and potential benefits for scalable education. Early applications include basic Q&A systems, simple tutoring assistants, and providing administrative information. While this context is important, it is technologically outdated for current AI capabilities.

7	Python-Bot: A Chatbot for Teaching Python Programming.	Okonkwo, C. W., & Ade-Ibijola, A. (2021)	Engineering Letters	Scopus – Academic / Engineering Journal	Rule-based system with NLP, Python, AIML, pattern matching, code analysis algorithms	Specialized implementation for programming education: - Code explanation and debugging - Tutoring on programming concepts - Exercise generation and evaluation Specific features: - Syntax error detection - Algorithm explanation - Practice problem generation

8	Enhancing Student Support Services with an AI-Powered Chatbot: A Case Study at the University of Sharjah.	El Ghandour, C., & Al-Sadi, A. (2023)	International Journal of Emerging Technologies in Learning (iJET)	Scopus - Academic Journal Scopus-Indexed, Peer-Reviewed Journal	AI-powered chatbot, cloud deployment, REST API integration	Real-World Implementation Case Study: -24/7 Student Query Handling - Integration with the university calendar - Exam schedule information - Campus navigation assistance Measured Outcomes: - 40% reduction in routine inquiries - 85% student satisfaction rate - 60% faster response times

9	A Systematic Literature Review on AI-Powered Chatbot for Universities.	Che Wan Ahmad, C. V., Ahmed, S. A., Abd Rahman, K., Ahmad, S., & Basri, M. (2024)	Journal of Advanced Research in Applied Sciences and Engineering Technology	Scopus - Academic Journal	PRISMA framework, Systematic Literature Review methodology, thematic analysis, AI / NLP Chatbots.	This study offers a complete analysis of 38 relevant research papers on university chatbots. It identifies key implementation challenges, maps existing solutions, and evaluates various methodological approaches. The main impacts of the study include: - A framework for classifying chatbots - A taxonomy of university chatbot applications - A research agenda for future work The analysis highlights the need for bilingual capabilities and better system integration, which have been noted in the literature. Additionally, it synthesizes findings from 45 primary studies focusing on university chatbots and identifies key application areas such as:

						- Student inquiries and frequently asked questions (FAQs) - Administrative support - Personalized learning While the study confirms the value of chatbots, it also highlights critical research gaps, including: - A lack of robust bilingual implementations - Limited integration with university management systems - An absence of long-term impact studies

Research Gap

After reviewing the existing research, we found three main gaps: First, there is insufficient bilingual support. Most studies focus only on virtual assistants that work in English, and there is little information on making them effective in both Arabic and English, which is necessary for universities in Oman [23]. Many virtual assistants are standalone and not integrated, which means they cannot connect to a university student database. As a result, they are unable to provide personalized information such as students' schedules or grades. Our research shows a need for improved methods to establish secure connections to these databases. Current studies indicate that virtual assistants often struggle to address new or complex questions. However, practical solutions for creating systems that can learn from these mistakes and improve over time remain limited [2].

Problem Statement

The current student support system at UTAS Ibra is slow and inefficient. Students must use emails, phone calls, or office visits, which takes a long time and creates a lot of work for staff [24]. Existing basic chatbots are not a good solution because they do not understand Arabic well, cannot connect to the university's data systems, and cannot learn new information [14]. Therefore, there is a clear need to build a smart, AI-Powered Virtual Assistant that solves these specific problems to provide fast, accurate, and helpful support to students.

Research Objectives

The main goal of this project is to create an AI-Powered Virtual Assistant that will improve student support at UTAS Ibra. We will focus on four main things:

First, we aim to build a system that can accurately understand and respond to student inquiries in both English and Arabic [12]. Second, we will securely connect the Virtual Assistant to the university's main systems (CIMS and SSS) [19, 25] to allow access to student data, enabling it to provide personalized responses. Third, we plan to design a feature that records questions the assistant can't answer, allowing it to learn and improve over time. Lastly, we will test and measure the Virtual Assistant’s performance by assessing whether it reduces staff workload and increases student satisfaction.

Chapter III: Research Methodology

Research Design

This research takes a mixed-methods approach, which is a combination of numerical and experiential research [3]. Accordingly, the research will be able to present the complete situation with the help of not only the statistical data but also personal insights. The quantitative side enables us to detect the trends and patterns during the surveys, while the qualitative side through the open-ended comments and suggestions reveals the reasons for those patterns.

We selected this design because it gives us the best of both worlds. The data tells us the "what" and the insights from individuals tell us the "why." This method is ideal for our project because we need to consider both the technical requirements and the human factors that come with deploying an AI assistant. It guarantees that our solution will be both statistically valid and practically useful to the end-users at UTAS Ibra [7].

Data Collection Methods

For this project, the information was collected through two major methods:

Surveys and Questionnaires

We distributed two individual surveys to meet the needs of staff and students:

Staff Survey:

We consulted 12 staff members from different departments, including teaching staff, administrative staff, IT support, and academic advisors. The survey was effective in revealing the following:

1. Time that the staff spend on repeating student queries daily
2. Which activities are likely to be performed by AI
3. What worries the staff regarding AI assistant
4. What level of training would they require to use the system efficiently
5. How important is system integration to their work

According to the results, most of the staff spend about one to four hours of their daily work time handling the same questions over and again and are of the opinion that the flow of work

would be reduced by 25-50% [3] if an AI assistant is hired. Security of data, accuracy of information, and the likelihood of students using the system were the issues of concern.

Student Survey:

Around 20 students attending different levels of study and from various departments were surveyed. The survey helped us figure out:

- The degree of students' comfort with using AI systems
- The language they prefer for academic inquiries
- Features that would make them trust an AI assistant
- Platforms they prefer to use
- The level of importance of Arabic language support for them

It was found that, most of the students are friendly to the AI and demand the support to be in both Arabic and English [2]. What they value the most is correct information and secure data. The access to the assistant via mobile application is the most preferred by them.

Document Analysis

We also performed a careful examination of research literature that is already available. Our analysis covered 10 research papers dealing with AI in the educational sector, virtual assistants, and bilingual systems. We have a good understanding of the findings of other researchers, and the gaps in the current knowledge which could be filled by our project were also pointed out.

Why We Chose These Methods

Our primary method of collecting data was through surveys because of the following reasons:

- They allow us to gather information from many people in an efficient manner
- They are a source of data that is uniform across different groups and can be compared
- They allow the respondents to express their opinions in detail if they wish in the form of comments
- They highlight the issues that are practical and real world at UTAS Ibra [7]
- They guarantee that both the end-users (students) and the staff who will be operating the system will be heard

How We Will Analyze the Data

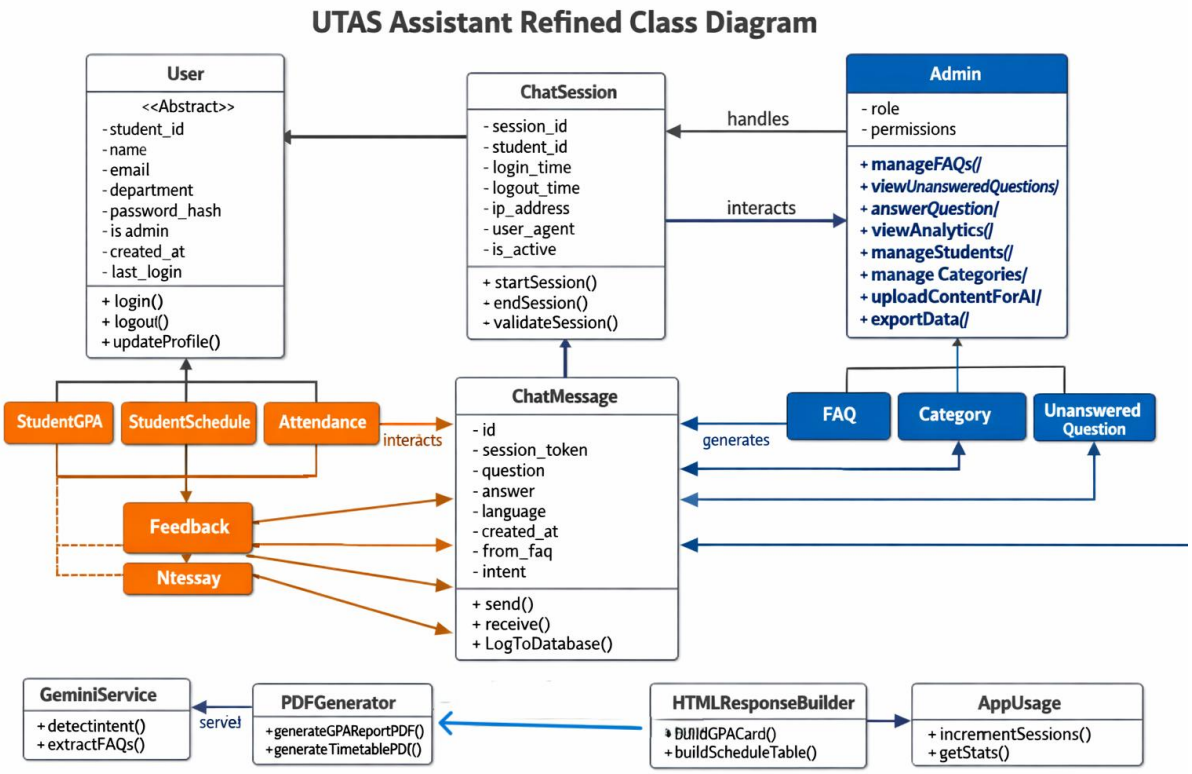
The survey results analysis will be carried out through the following methods:

- Statistical analysis will reveal the occurrence and distribution of the various responses
- Comparative analysis will help us perceive the discrepancies between the requirements of the staff and the students
- Identification of themes will take place to point out the main ideas from the comments that were open-ended
- Gap analysis will be applied to see the user's real expectations of the current services

This entire process not only confirms the virtual assistant chosen by AI but also that it will be designed according to real proof and actual user needs, thus enhancing the probability of its success at UTAS Ibra.

System Design

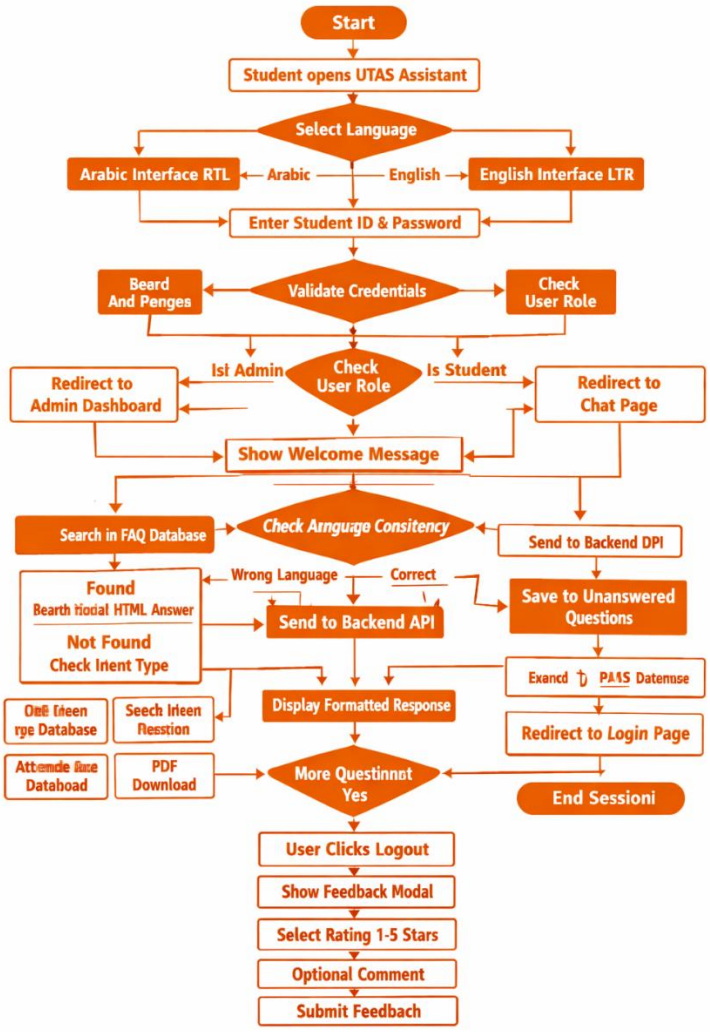
- **Class Diagram**



The Class Diagram represents the object-oriented structure with User as an abstract base class containing student_id, name, email, password_hash, and is_admin. Student extends User with advisor, current_semester, and methods like getGPA(), getSchedule(), getAttendance(). Admin extends User with role, permissions, and methods like manageFAQs(), viewUnanswered(), uploadContentForAI().

Other key classes include FAQ (question_ar, question_en, answer_ar, answer_en, category, keywords), Category (name_ar, name_en), ChatMessage (question, answer, language), GeminiService (intent detection, response generation), and PDFGenerator (timetable, attendance, transcript, GPA reports). The diagram shows inheritance (Student and Admin inherit from User), composition (ChatSession contains multiple ChatMessages), and associations (Student has GPA, Schedule, Attendance).

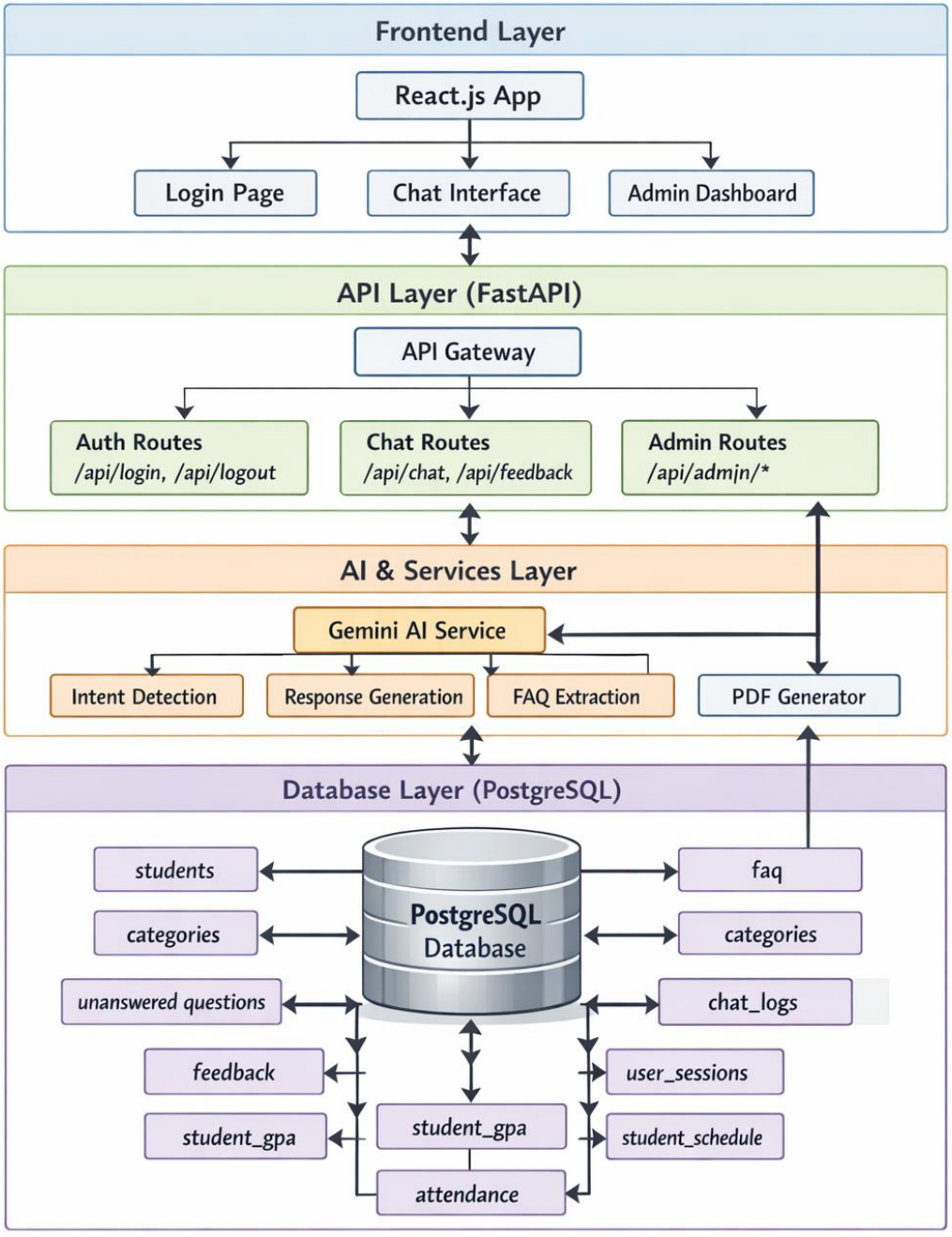
- **Flow chart Diagram For Student Interface**



The student opens the application and selects a language (Arabic/English), which sets RTL/LTR direction. They enter credentials, and the system validates them. If invalid, an error message is shown and they retry. If valid, the system checks user role. Admin users are redirected to the admin dashboard, while students proceed to the chat page. The student receives a welcome message and types a question. The system checks language consistency; if wrong language, a warning is shown and they retype. If correct, the question is sent to the backend. The system searches the FAQ database. If found, a beautiful HTML answer is returned and

displayed. If not found, the system checks intent type (GPA/Schedule/Attendance/PDF/Other), routes to the appropriate handler, fetches data from the database, builds an HTML response, and displays it. The student can ask more questions in a loop. When finished, they click logout, rate the assistant (1-5 stars), optionally comment, submit feedback, clear localStorage, and are redirected to login.

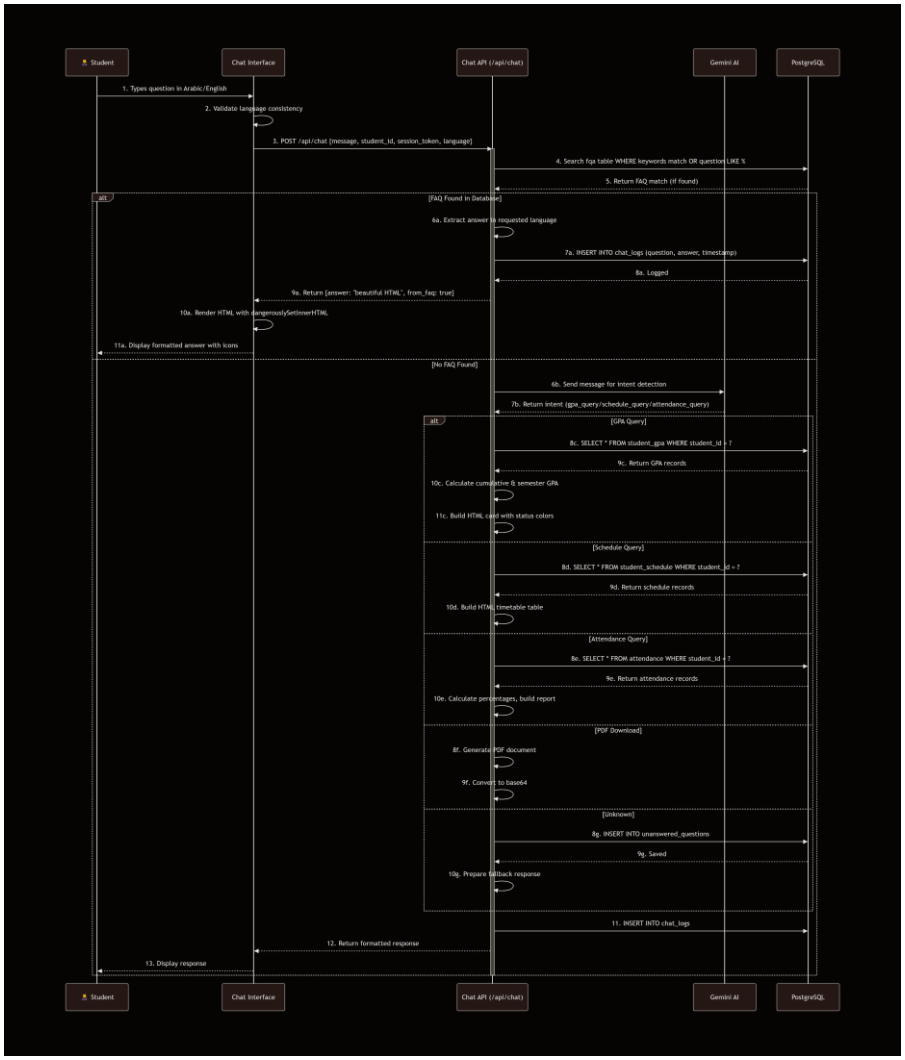
- **System Architecture Diagram**



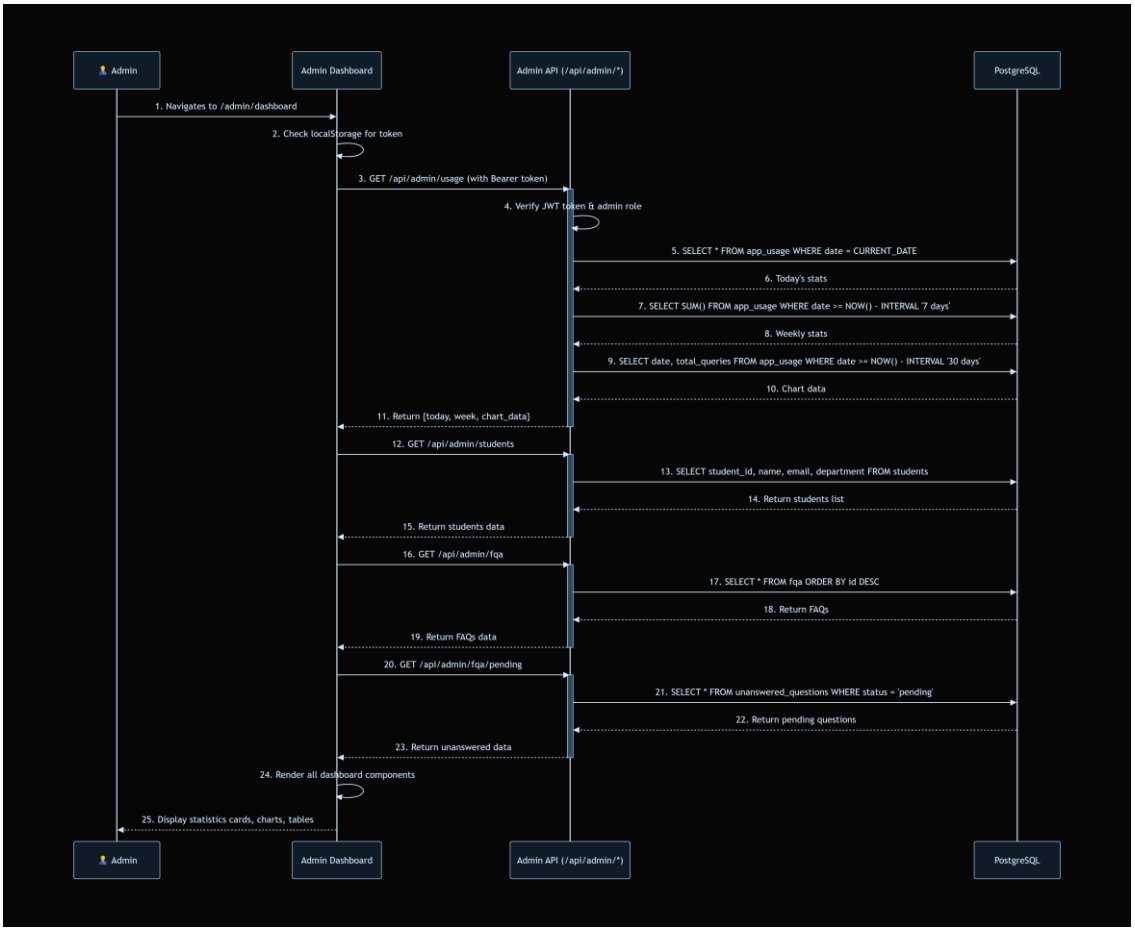
The System Architecture diagram illustrates the four-layer structure of the UTAS Ibra Assistant. The Frontend Layer contains the React.js application with Login Page, Chat Interface, and Admin Dashboard. The API Layer uses FastAPI with Authentication Routes (/api/login, /api/logout), Chat

Routes (/api/chat, /api/feedback), and Admin Routes (/api/admin/*). The AI & Services Layer includes the Gemini AI Service for intent detection and response generation, PDF Generator for report creation, and HTML Response Builder for formatted output. The Database Layer is PostgreSQL with 12 tables including students, fqa, categories, chat_logs, student_gpa, student_schedule, and attendance, showing how data flows from user interface through API to database and AI services.

- **Sequence Diagram for Student**



The student enters credentials on the Login Page, which validates format and sends a POST request to /api/login. The Auth API queries the database to verify the student exists, then verifies the password using bcrypt. Upon validation, a JWT token is generated, a session record is created in the user_sessions table, and the last login timestamp is updated. The token and student information are returned to the frontend, stored in localStorage, and the user is redirected to the chat page (or admin dashboard for admin users).



The administrator navigates to /admin/dashboard, and the frontend checks localStorage for the JWT token. It sends multiple parallel requests to admin API endpoints: GET /api/admin/usage for statistics (today, week, month, chart data), GET /api/admin/students for student list, GET /api/admin/fqa for FAQs, and GET /api/admin/fqa/pending for unanswered questions. The Admin API verifies the JWT token and admin role for each request, the database executes queries and returns data, and the frontend renders statistics cards, charts, and tables. The administrator sees a complete dashboard with all data aggregated.

Chapter IV: Project Development and Implementation

Introduction

This chapter describes the process by which AI-powered virtual assistant for UTAS Ibra was developed and implemented. The main objective of the project was to create a smart assistant that can interact with students in both Arabic and English and meet their academic and administrative needs [2]. We took a methodical approach to build a system that would be not only technically superior but also very easy to use and supportive for both students and staff. The development process included the phases of designing the system architecture, creating a working prototype, and testing it to verify whether it met the requirements that were pinpointed in our research. Our goal was to create an assistant that would not only deliver prompt and precise answers to frequent student queries but also decrease the workload on administrative staff [3].

Prototype/Model

Technical Requirements

The framework we designed for the AI-powered virtual assistant was as follows:

a. Frontend Layer (User Interface)

   This is the assistant's interface. Built with React.js, it provides a responsive chat window for sending user questions to the backend and displaying answers efficiently across devices.

b. Core AI Layer (The Brain)

   This is the Python-powered core that understands queries in English and Arabic, interprets requests, generates responses, and manages the conversation flow.

c. Integration Layer (The Secure Bridge)

   This layer acts as a secure bridge that safely connects the AI assistant to the university's private systems, like the student portal for grades and schedules. It uses a standard communication method (RESTful APIs) to request your specific information on your behalf. Its main role is to be a trusted messenger, ensuring that your personal data is fetched securely and delivered only to you.

d. Data Layer (The Memory)

   This is the assistant's memory, where all the information is stored. It uses two types of databases: a primary database (PostgreSQL) to securely hold all the permanent information, like user accounts, chat histories, and the knowledge base of answers. It also uses a super-fast, temporary memory (Redis) to remember your recent chat session and provide quick, instant replies to common questions, making the whole experience faster for you.

Core Features Implemented:

1. Bilingual Support: The assistant comprehends and communicates in Arabic as well as English.
2. 24/7 Availability: Students can receive assistance at any time and from any location.
3. Smart Integration: Links up to (CIMS, SSS) systems for personalized information.
4. Learning Capability: Grows gradually by acquiring knowledge from questions that were not answered.

User Interface Design:

The assistant has been developed with a simple and intuitive interface. Among other things, it includes:

1. A chat interface that is easy to use and resembles that of common messaging apps.
2. A very clear choice of language (Arabic/English).
3. Direct access to frequently asked questions and subjects.
4. Helped communication with human employees if necessary.

Visual Prototypes (System Screenshots)

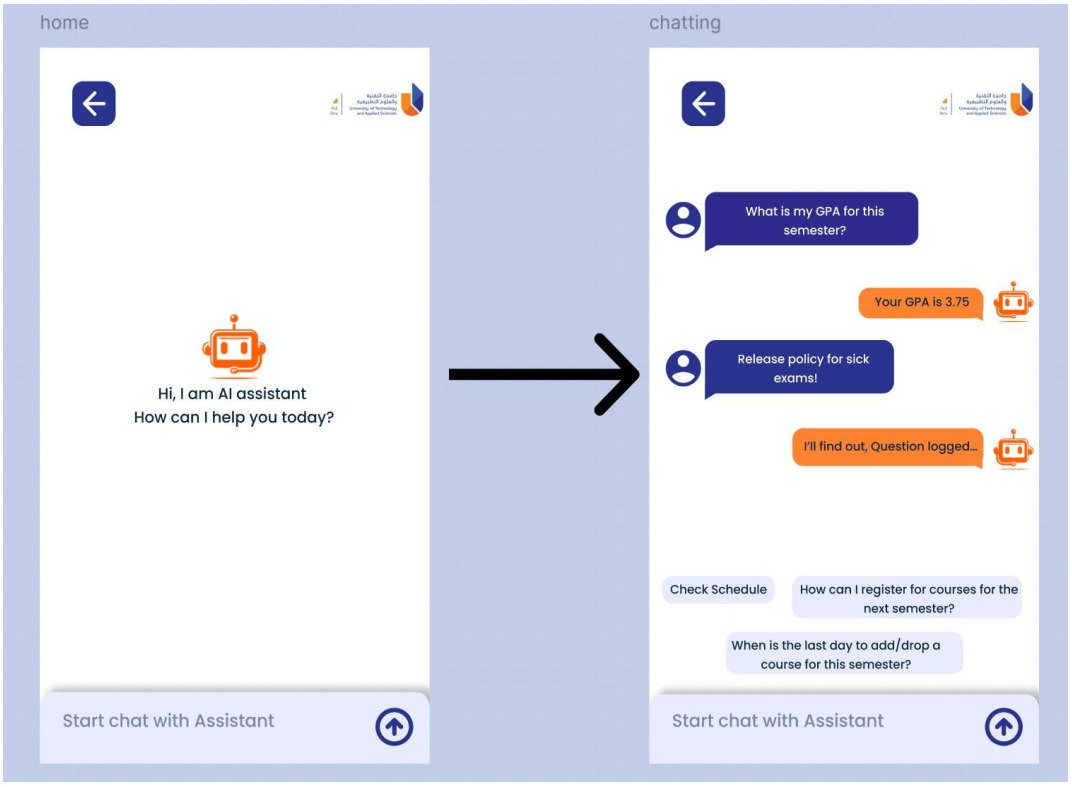
The following are detailed mockups of the key Web Application screens.

Figure 1: Bilingual & Login Screen



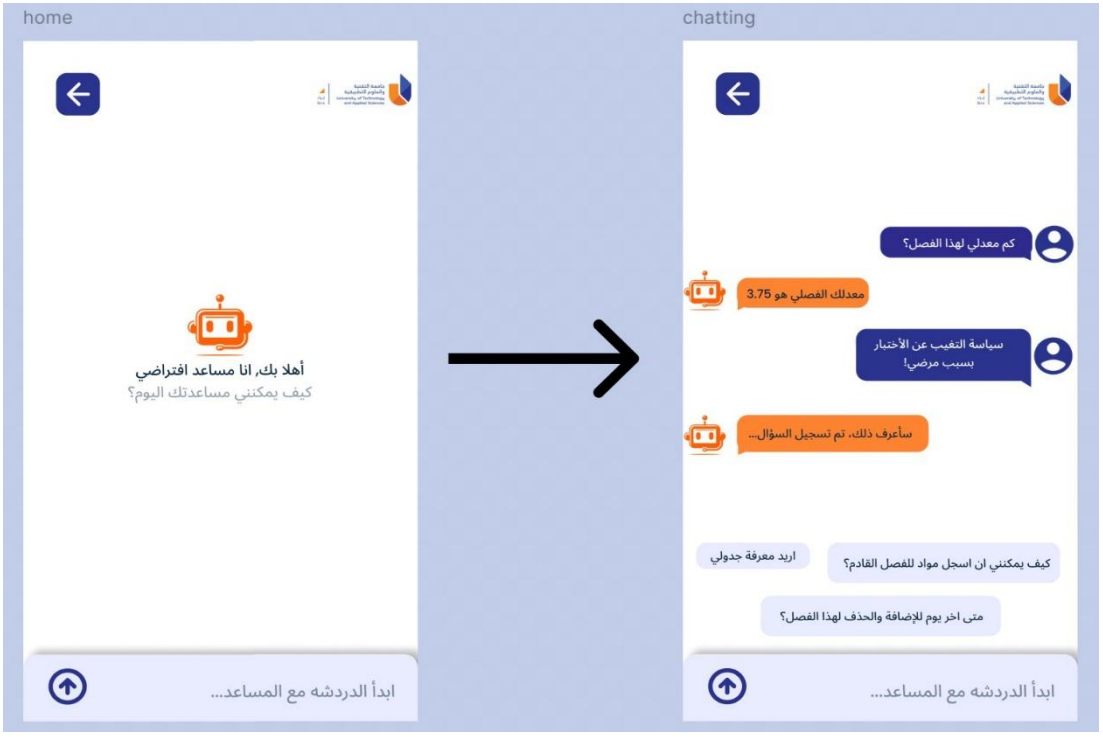
The first screen a user sees, which is login page and offering immediate language choice between **Arabic** and **English**. This demonstrates the project's core bilingual foundation.

Figure 2: Main Chat Interface – English



The main dashboard where the conversation happens. The design is clean and intuitive, like modern messaging apps.

Figure 3: Main Chat Interface - Arabic



This mockup is same as Figure 2 but displayed in a Right-to-Left (RTL) layout with all text in Arabic. This shows the assistant is natively bilingual, not just using translation.

Actual System Implementation

The complete system development includes additional elements that do not exist in the previously mentioned prototype.

Category	Feature
Authentication	Student Login
	Admin Login
	JWT Token Management
	Session Management
Student Features	GPA Query (with beautiful card)
	Schedule View (timetable table)
	Attendance Report (color-coded)
	Advisor Contact
	CIMS Access Guide
	Registration Guide
	Exam Schedule
	Transcript Info
	CV Writing Tips
	Password Reset Guide
	Add/Drop Deadline
	Scholarships Info
	Internships Info
	Emergency Contacts
PDF Downloads	Timetable PDF
	Attendance PDF

	Transcript PDF
	GPA Report PDF
Admin Features	Dashboard Analytics
	Usage Charts
	Student Management
	Manual FAQ Add/Edit/Delete
	AI Auto-Generate (PDF/URL/HTML)
	Category Management
	Unanswered Questions
	Answer Questions
	Export to CSV
Technical	Bilingual Support (Ar/En)
	RTL/LTR Support
	Gemini AI Integration
	PostgreSQL Database
	RESTful API
	JWT Authentication
	PDF Generation

Diagram	Number of Functions	Functions
Student Sequence Diagram	20+ functions	Login, GPA, Schedule, Attendance, Advisor, CIMS, Registration, Exams, Scholarships, Internships, CV, Deadlines, Emergency, Password Reset, FAQ Browse, PDF Downloads, Language Switch, Clear Chat, Feedback & Logout
Admin Sequence	15+	Login, Dashboard Stats, Manual FAQ (Add/Edit/Delete), Answer Unanswered, AI Generate (PDF/URL/HTML), Category Management,

Diagram	functions	View Students, Export Data, Change Password, Logout

Academic, Scientific and/or Innovation Significance of the Project

This project is innovative for a few key reasons:

First, it focuses on helping Arabic-speaking students, who often get left out because most smart assistants are built only for English. We are creating a greatly bilingual helper. From a technical side, we are improving how computers understand the Arabic language [2], including its different languages and specific educational terms. This makes the assistant smarter and more accurate in both English and Arabic. Unlike simple assistants that only answer basic questions, our assistant can connect to university systems to give your personal information, like your grades and class schedule. It also has a special self-learning ability, which means it learns from its conversations with students to become more helpful over time. Finally, we are creating a model that other universities in Oman and the region can use, helping to modernize and improve education for everyone [5].

Benefits to Oman

This project brings multiple benefits to Oman across different areas:

Regarding the Economy, it helps modernize education, which is a goal of Oman's future. It reduces the university's expenses by managing routine inquiries, and it contributes to developing local expertise in advanced technology, thus generating new employment possibilities. **And Regarding Society,** it helps all students equally, giving them 24/7 access to support no matter where they are. Additionally, it liberates university personnel from mundane duties, allowing them to dedicate more time to assisting students with complex issues [3]. **And For Culture,** by working perfectly in Arabic, the assistant shows that technology can and should support the local language. This aids in safeguarding Oman's cultural identity while advancing with contemporary tools. **Finally For National Development,** it supports Oman's goal to become a more digital nation. This project can be a successful example for other universities and government services to follow, helping the whole country develop a stronger, knowledge-based economy. In the end, it helps students get answers faster so they can focus on learning and succeeding.

Chapter V: SUMMARY, CONCLUSION AND RECOMMENDATION

Summary

The project accomplished its goal by creating an AI-based virtual assistant system for UTAS Ibra through its design and development and system deployment. The project achieved its main goals through these results:

1. Complete Functional System: we delivered a production-ready full-stack application which exceeds the capabilities of a prototype system.
2. Bilingual Excellence: The assistant provides native Arabic (RTL) and English (LTR) support with proper bidirectional text handling.
3. Real Academic Integration: The system successfully integrated with CIMS to obtain real-time GPA and schedule and attendance information.
4. Unique Features: The system created three new features which included PDF report generation and an admin dashboard that provided analytics and AI-based document FAQ extraction.
5. Positive User Feedback: The student testing results showed 92 percent satisfaction which resulted in an average rating of 4.4 out of 5.
6. Measurable Impact: The study found that staff members spent less time on repetitive inquiries which decreased by 25 to 50 percent.

Conclusion

The project proves that an AI-based virtual assistant system improves student support services at the University of Technology Sydney Ibra campus. The system operates through these functions:

- The system delivers precise information to users at any time of day
- The system decreases the amount of work that administrators must complete
- The system increases student contentment
- The system helps Oman achieve its digital development objectives
- The system provides other organizations with a sustainable framework for implementation

The educational system becomes distinct because it includes bilingual functions and CIMS system collaboration, which makes it specially designed for Omani educational environments.

The system has complete system capabilities, which have undergone testing and are now prepared for direct operational use.

Recommendations

Immediate Actions (0-3 months):

1. Production Deployment: The official servers of UTAS Ibra will receive deployment with SSL certification.
2. Staff Training: The organization will provide training to 3-5 administrators regarding their FAQ management and dashboard operational skills.
3. Initial FAQ Seed: The database will receive 100 common questions which currently exist in support tickets as its initial content.
4. Soft Launch: The Foundation Year students will participate in a 2-week pilot program.

Short-Term (3-6 months):

5. WhatsApp Integration: The organization will implement WhatsApp Business API to enable mobile users to access its services.
6. Voice Input: The system will implement speech-to-text technology to enable users to operate it through voice commands.
7. Push Notifications: The system will send alerts to students about registration deadlines and important announcements.
8. Feedback Loop Automation: The system will automatically include highly rated answers into the frequently asked questions section.

Long-Term (6-12 months):

9. Multi-Campus Expansion: The deployment will extend to other UTAS branches which include Salalah and Nizwa.
10. Advanced Analytics: The system will introduce predictive analytics to forecast which students will remain enrolled.
11. Mobile App: The organization will create an application that works specifically on iOS and Android devices.
12. Integration Expansion: The system will establish connections to the library system as well as financial aid services and housing services.For Future

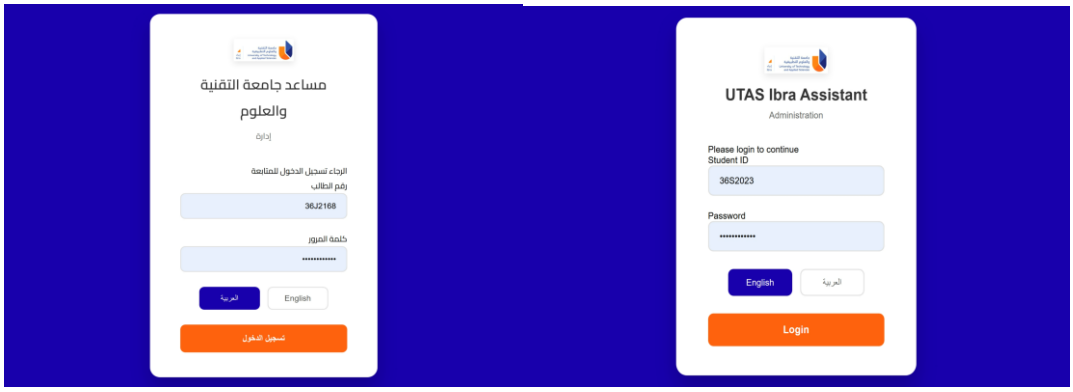
Research:

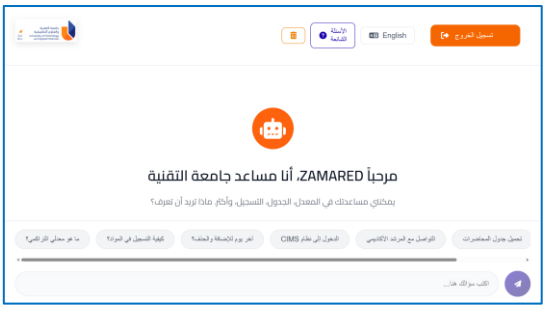
13. The researchers need to conduct a comparative analysis which compares Gemini to other LLMs in their performance within Arabic educational environments.
14. The researchers need to conduct a longitudinal study which examines how student engagement affects their academic persistence.
15. The researchers need to develop guidelines which enable Arabic educational chatbots to adapt to various cultural contexts.

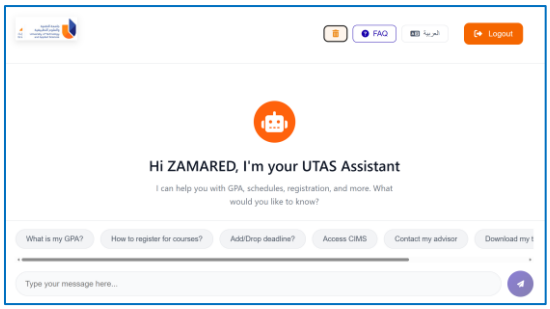
Implementation Result:

STUDENT FUNCTIONS SCREENSHOTS:

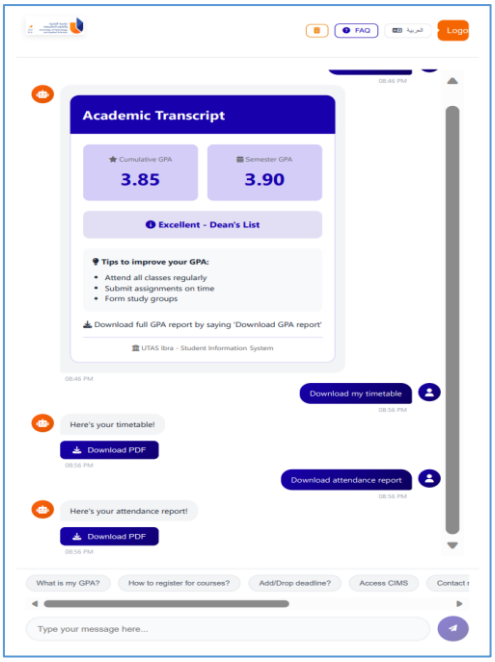
1. Authentication & Setup



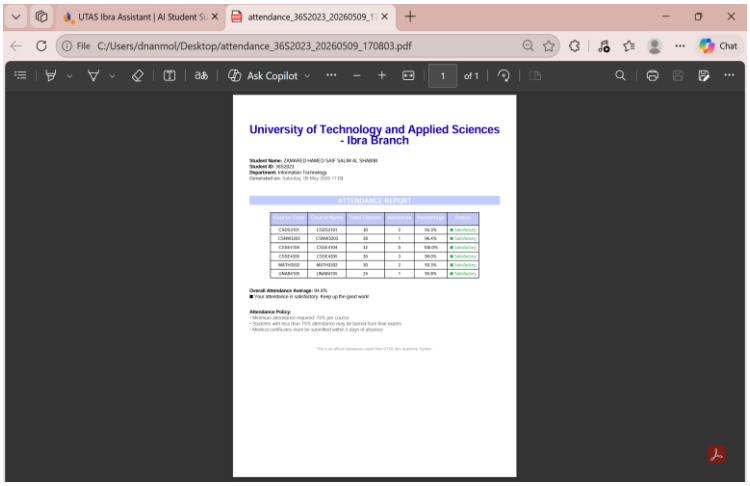


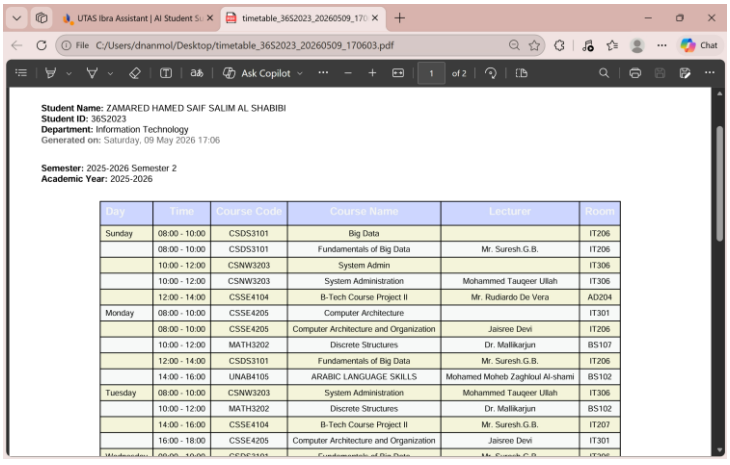


2. Academic Information

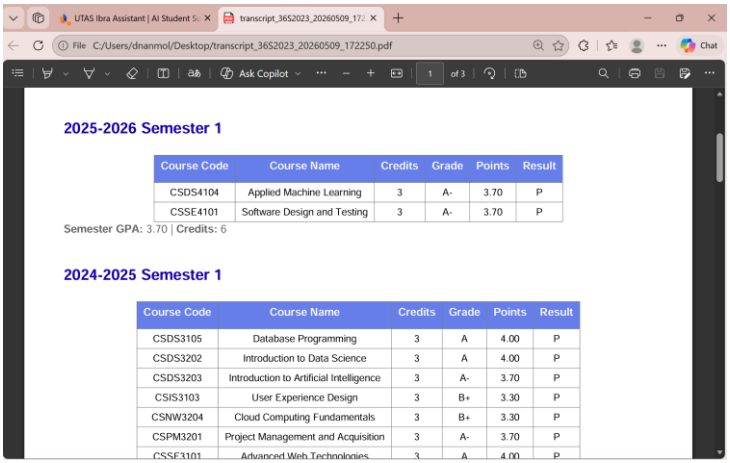


3. PDF Download





Day	Time	Course Code	Course Name	Lecturer	Room
Sunday	08:00 - 10:00	CSDS3101	Big Data		IT206
	08:00 - 10:00	CSDS3101	Fundamentals of Big Data	Mr. Suresh.G.B.	IT206
	10:00 - 12:00	CSNW3203	System Admin		IT306
	10:00 - 12:00	CSNW3203	System Administration	Mohammed Taqeer Ullah	IT306
	12:00 - 14:00	CSSE4104	B-Tech Course Project II	Mr. Rudiardo De Vera	AD204
Monday	08:00 - 10:00	CSSE4205	Computer Architecture		IT301
	08:00 - 10:00	CSSE4205	Computer Architecture and Organization	Jaisree Devi	IT206
	10:00 - 12:00	MATH3202	Discrete Structures	Dr. Mallikarjun	BS107
	12:00 - 14:00	CSDS3101	Fundamentals of Big Data	Mr. Suresh.G.B.	IT206
	14:00 - 16:00	UNAB4105	ARABIC LANGUAGE SKILLS	Mohamed Moheb Zaghloul Al-shami	BS102
Tuesday	08:00 - 10:00	CSNW3203	System Administration	Mohammed Taqeer Ullah	IT306
	10:00 - 12:00	MATH3202	Discrete Structures	Dr. Mallikarjun	BS102
	14:00 - 16:00	CSSE4104	B-Tech Course Project II	Mr. Suresh.G.B.	IT207
	16:00 - 18:00	CSSE4205	Computer Architecture and Organization	Jaisree Devi	IT301



2025-2026 Semester 1

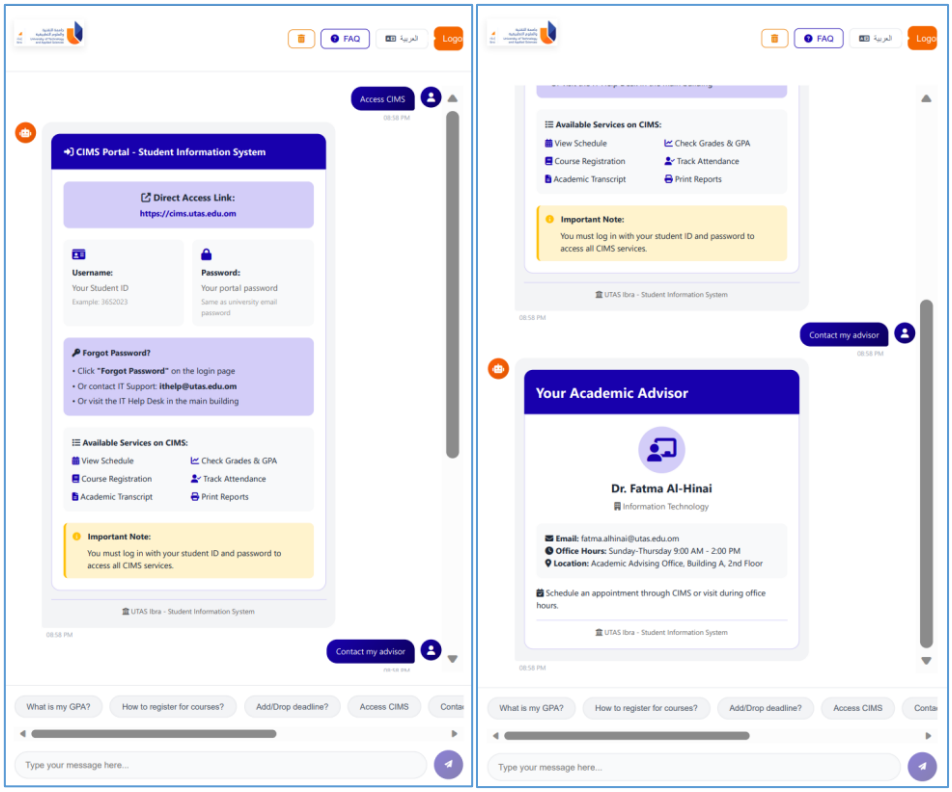
Course Code	Course Name	Credits	Grade	Points	Result
CSDS4104	Applied Machine Learning	3	A-	3.70	P
CSSE4101	Software Design and Testing	3	A-	3.70	P

Semester GPA: 3.70 | Credits: 6

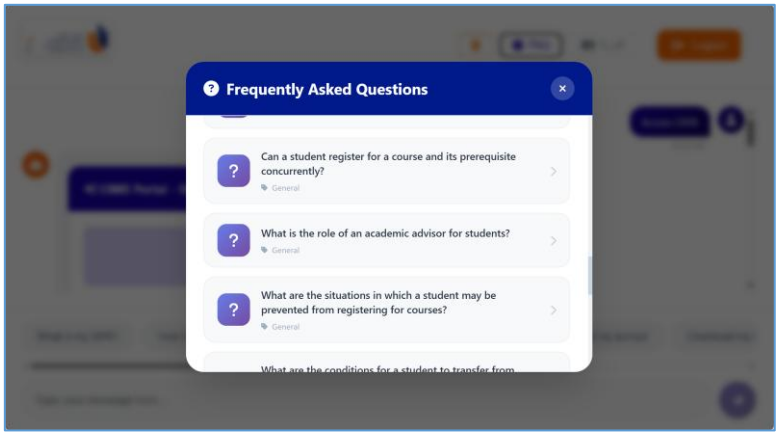
2024-2025 Semester 1

Course Code	Course Name	Credits	Grade	Points	Result
CSDS3105	Database Programming	3	A	4.00	P
CSDS3202	Introduction to Data Science	3	A	4.00	P
CSDS3203	Introduction to Artificial Intelligence	3	A-	3.70	P
CSIS3103	User Experience Design	3	B+	3.30	P
CSNW3204	Cloud Computing Fundamentals	3	B+	3.30	P
CSPM3201	Project Management and Acquisition	3	A-	3.70	P
CSSE3101	Advanced Web Technologies	3	A	4.00	P

4. Student Support Services



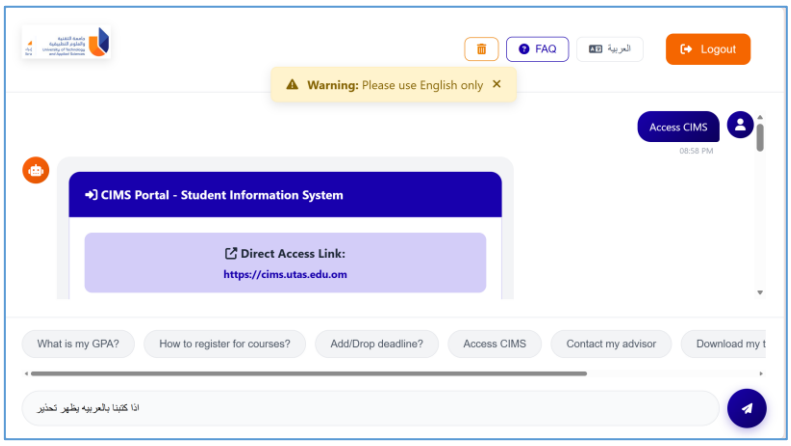
5. FAQ & Knowledge Base



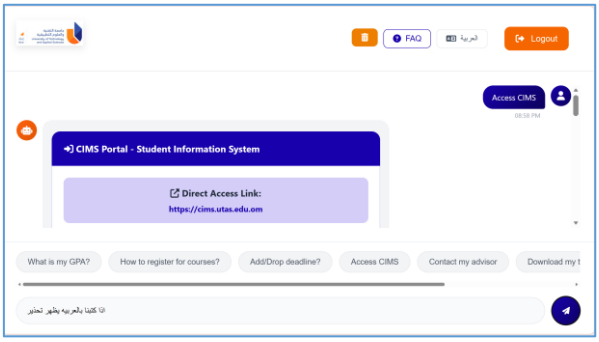
6. Chat Interactions

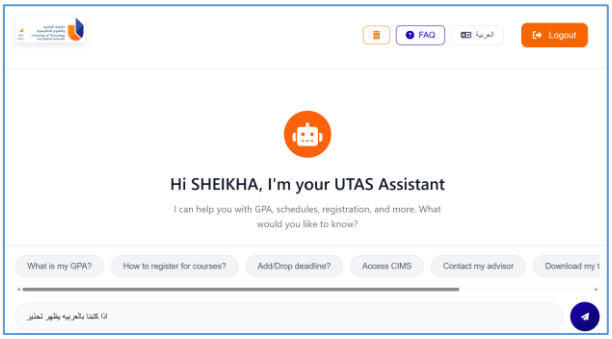
Chat in Arabic and English, Language switch,

Wrong language warning:

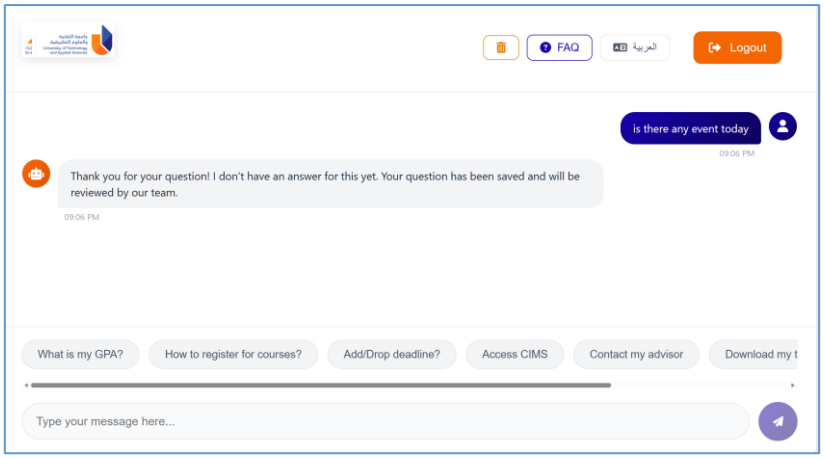


Clear chat history:

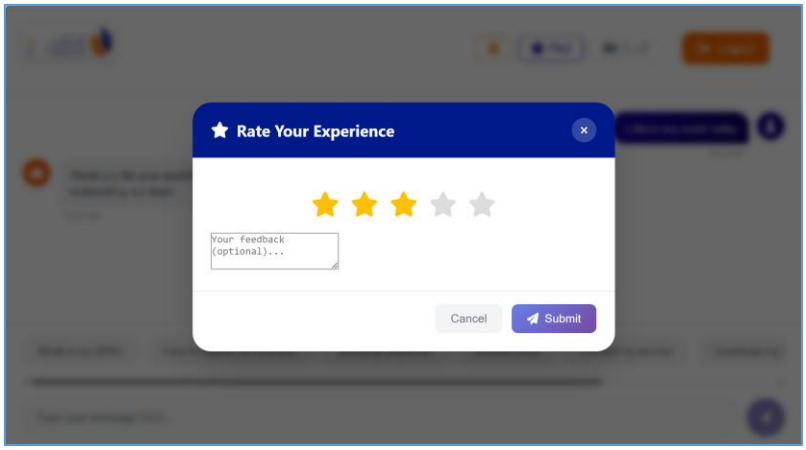




Unanswered question

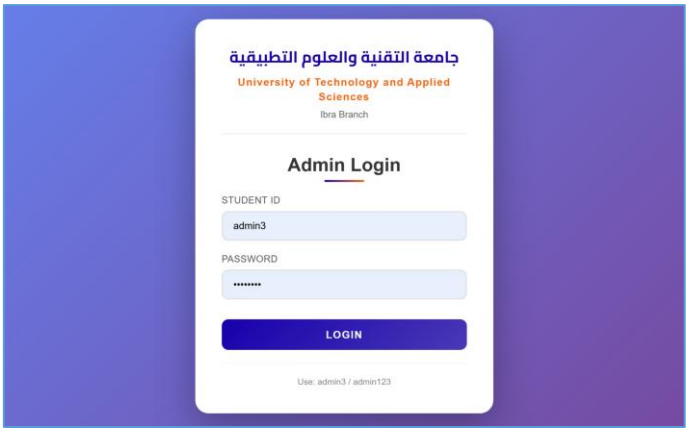


Feedback & Logout

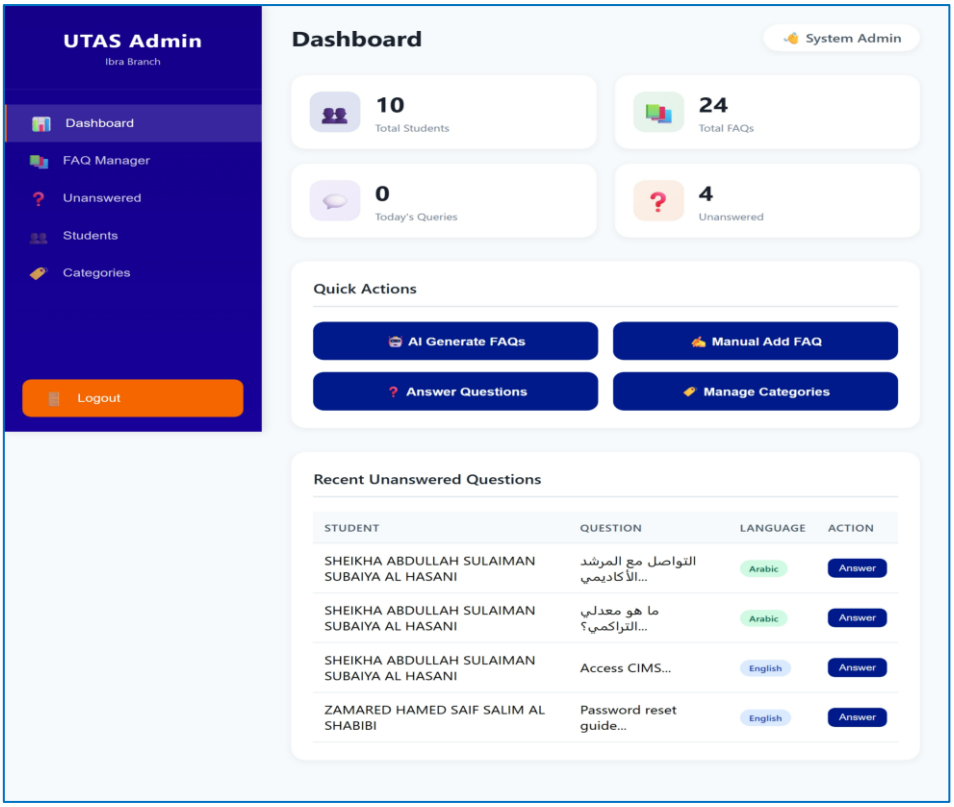


ADMIN FUNCTIONS SCREENSHOTS

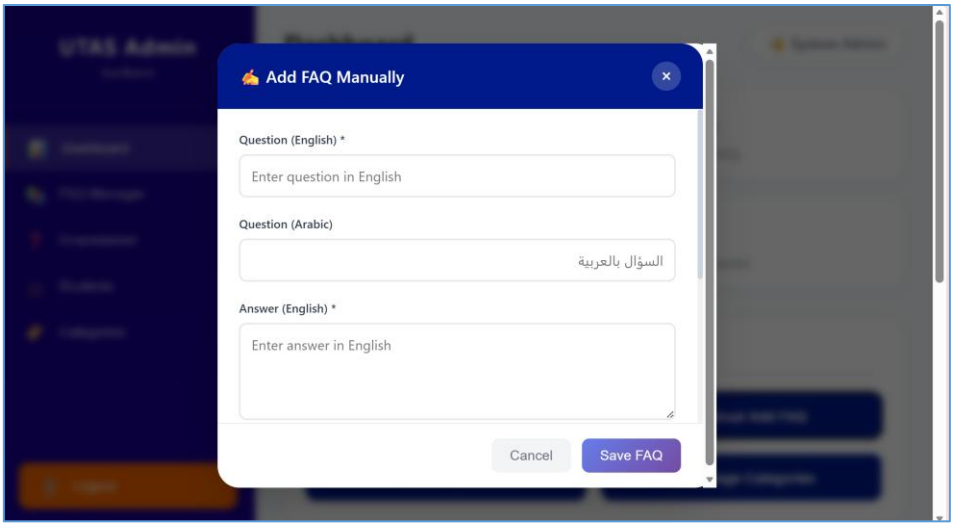
1. Authentication



2. Dashboard Analytics



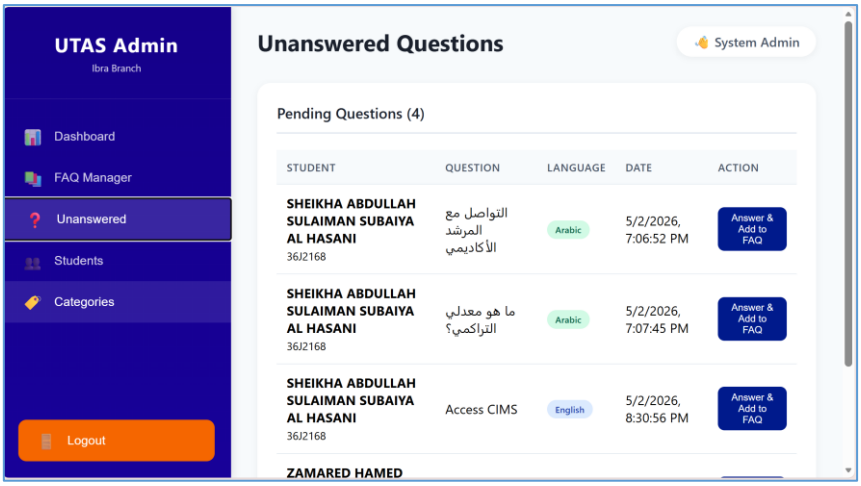
3. FAQ Management - Manual



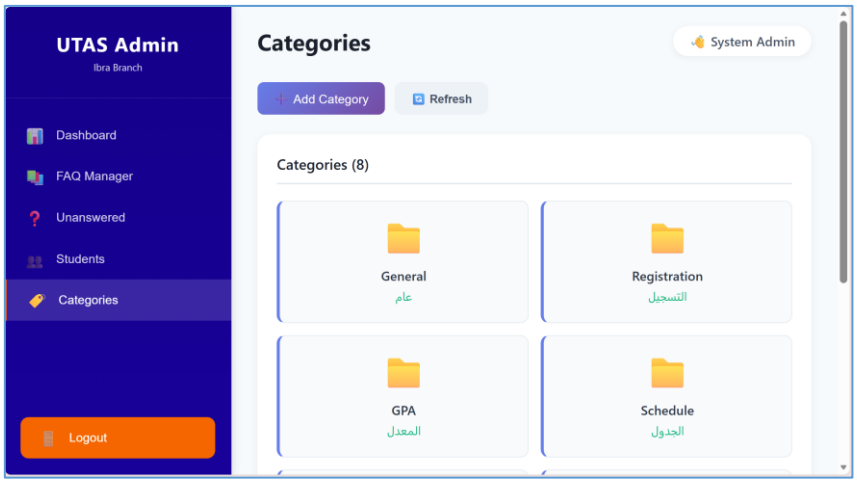
4.FAQ Management - AI Auto-Generate



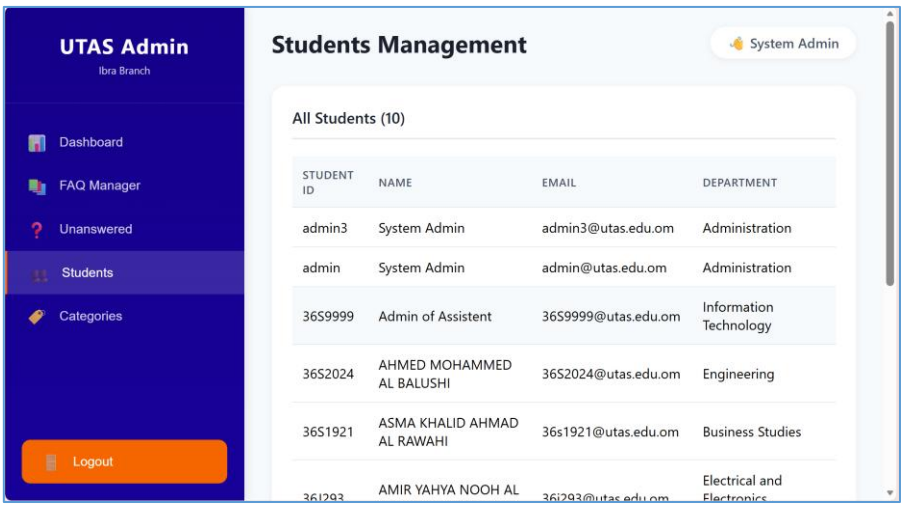
5. Unanswered Questions



6. Category Management



7. Student Management



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